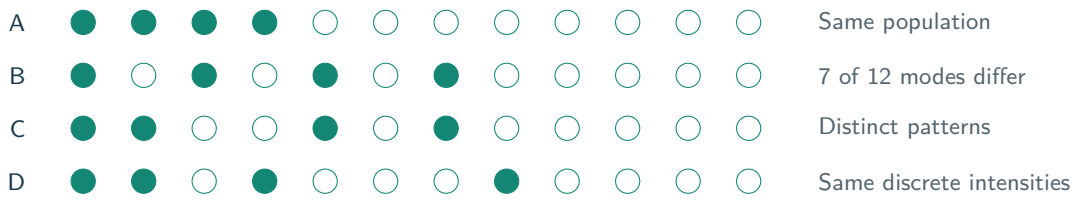
One external geometry test. An inspectable numerical foundation.

The generic representation preserves the declared diffraction amplitudes.
Six synthetic transformations, four occupation fixtures and one public titanate CIF have been evaluated. Measured structure identification remains unassessed.

Generic core	Explicit sites, species, occupancies, row-basis cells, integer supercells, source/tile mappings, exact species counts and supplied periodic exclusion witnesses. **47 implementation tests pass.**
Independent controls	Rational cosets and scalar complex sums check six transforms. Four ring fixtures test equal means, different correlations, periodic exclusions and ambiguity. **57 checks pass; two are not applicable.**
External CIF	A21am layered titanate: **60 site/species records**, including mixed Ca/Sr alternatives; **48 occupancy-weighted atoms**. Literal composition and CIF round trips retained.
Forward agreement	Across **2,197 reciprocal vectors**, parent amplitudes agree with Gemmi's separate calculator to a maximum absolute error of 4.08×10^{-13}. Two nontrivial supercells pass complete declared grids.

Scope of that agreement. The source contains no ADPs. Calculations deliberately use zero ADPs, neutral IT92 X-ray factors and no anomalous terms. Gemmi supplies symmetry admission and the common factor tables; its calculator is independently implemented, but the crystallographic inputs are shared. No fitted scale or background repairs are used.

The occupation controls also preserve ambiguity



Each row occupies four positions on a periodic twelve-site ring. Exhaustive integer enumeration of all **495** configurations confirms C/D share cyclic autocorrelation but are not related by translation or reversal. This is a discrete periodic mathematical result, not a chemical or powder-uniqueness claim.

A correction preserved, and a claim narrowed

Run001 passed its finite controls. Review then exposed large-coordinate numerical errors outside those fixtures. Its exact source/result remain archived. Version 0.1.1 corrects phase evaluation and expansion and passes revalidation. Under the published protocol, **VT07 is now development data for this revision**; a fresh material is needed for the next held-out test. Neither run was blinded discovery.

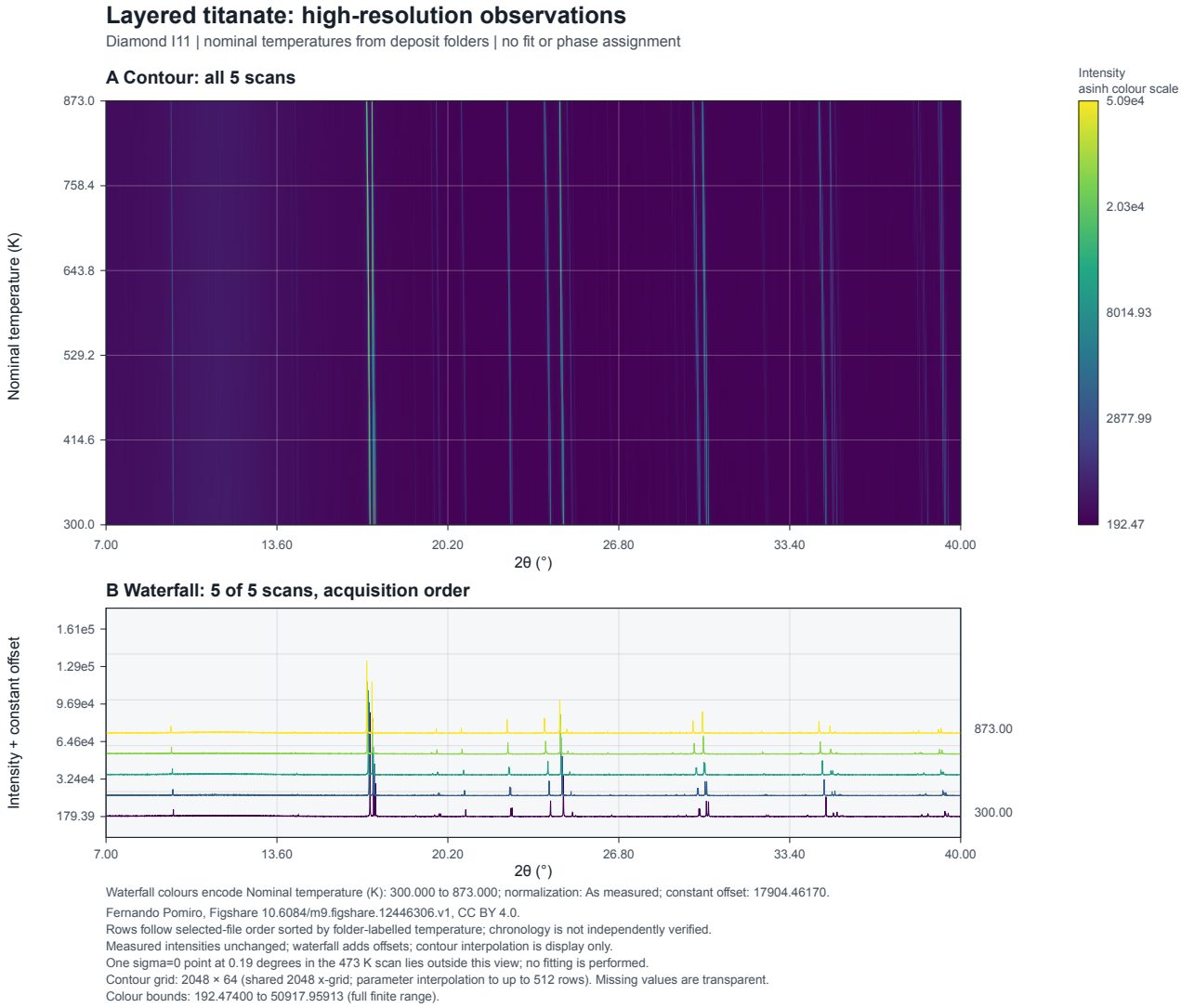
What remains before an experimental claim

The deposited TOPAS input is a different P21nm multiphase model and references an absent XYE filename. Full-profile reproduction needs a matched model, observations, ADPs and response. Next: admit a new occupational family, freeze the corrected core, then evaluate. Fe_7S_8 is a candidate whose structure/instrument inputs remain unresolved.

Code, protocol, amendments and numerical receipts / Gemmi scattering conventions. Editable sources and exact environment are included. No private structures or legacy private code are in this package.

The measurements available for the next test

Five deposited Diamond I11 observations, 75,000 points each. This is a source-data view, not a fit or a phase assignment. Temperatures are taken from deposit folder labels.



Rendered with Rust Powder Plot

Original intensities are preserved. Waterfall offsets and contour interpolation are display operations. The renderer's fixed "acquisition order" heading denotes the selected file order here; chronology has not been independently verified. One zero-sigma point at 0.19° in the 473 K scan lies outside this view and is not admitted to a weighted fit.

Source: Fernando Pomiro (2020), *Cif_files_and_exp_data.zip*, Figshare DOI 10.6084/m9.figshare.12446306.v1, CC BY 4.0. Native SVG, PDF/PNG conversions, input identities and a regeneration command are supplied. The plotted observations are distinct from the fixed-model amplitude controls on page 1.